

- Business Requirements Document (BRD)**

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- Project Overview**

The project involves designing and implementing a data processing pipeline to read customer and order data from CSV files, perform data cleansing and transformation, and load the data into a Delta table. The pipeline will also create and maintain summary tables to support business intelligence and analytics.

- Business Requirements**

1. Data Ingestion

- Read source CSV data from a specified volume location (`/Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/customerdata`` and `/Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/orderdata``) into Delta tables (``customer`` and ``order``).

2. Data Schema

- The schema for the ``customer`` table is: ``CustId``, ``Name``, ``EmailId``, ``Region``.
- The schema for the ``order`` table is: ``OrderId``, ``ItemName``, ``PricePerUnit``, ``Qty``, ``Date``, ``CustId``.

3. Data Cleansing

- Remove records with null values from both ``customer`` and ``order`` tables.
- Remove duplicate records from both ``customer`` and ``order`` tables.

4. Order Summary Table Creation

- Create a new table ``ordersummary`` in the catalog ``gen_ai_poc_databrickscoe`` and schema ``sdlc_wizard`` if it does not exist.
- The schema for the ``ordersummary`` table is: ``CustId``, ``Name``, ``EmailId``, ``Region``, ``OrderId``, ``ItemName``, ``PricePerUnit``, ``Qty``, ``Date``.

5. Data Transformation and Loading

- Join ``customer`` and ``order`` data using the ``CustId`` field.
- Load the joined data into the ``ordersummary`` table, which is a Slowly Changing Dimension (SCD) Type 2 table.

6. SCD Type 2 Table Maintenance

- Update the ``ordersummary`` table whenever there is a change in the ``customer`` table.
- Mark old records as Inactive and new records as Active in the ``ordersummary`` table.
- Update ``StartDate`` and ``EndDate`` accordingly.

7. Customer Aggregate Spend Table Creation

- Create a new table `customeraggregatespend` in the catalog `gen\_ai\_poc\_databrickscoe` and schema `sdic\_wizard` if it does not exist.
- The schema for the `customeraggregatespend` table is: `Name`, `TotalAmount`, `Date`.

8. Data Aggregation and Loading

- Aggregate the `TotalAmount` column from the `ordersummary` table, grouped by `Name` and `Date` columns.
- Load the aggregated data into the `customeraggregatespend` table.

- Functional Requirements**

1. The system shall be able to read CSV data from a specified volume location.
2. The system shall be able to perform data cleansing (remove null and duplicate records).
3. The system shall be able to create and maintain Delta tables (`customer`, `order`, `ordersummary`, and `customeraggregatespend`).
4. The system shall be able to perform data transformation (join `customer` and `order` data).
5. The system shall be able to maintain an SCD Type 2 table (`ordersummary`).
6. The system shall be able to aggregate data from the `ordersummary` table and load it into the `customeraggregatespend` table.

- Non-Functional Requirements**

1. The system shall ensure data consistency and integrity.
2. The system shall ensure data is processed in a timely and efficient manner.
3. The system shall be able to handle changes in the `customer` table and update the `ordersummary` table accordingly.

- Assumptions and Dependencies**

1. The input CSV files are in the correct format and location.
2. The Delta tables are created with the correct schema.
3. The system has the necessary permissions and access to the volume location and catalog.

- Success Criteria**

1. The data is successfully ingested from the CSV files into the Delta tables.
2. The data is cleansed and transformed correctly.

3. The `ordersummary` table is maintained correctly as an SCD Type 2 table.

4. The `customeraggregatespend` table is populated correctly with aggregated data.

- Risks and Mitigants**

1. Data quality issues: Implement data validation and cleansing checks to mitigate this risk.

2. System performance issues: Optimize the data processing pipeline and ensure sufficient resources are allocated to mitigate this risk.

By following this BRD, the project team should be able to design and implement a data processing pipeline that meets the business requirements and provides a robust and scalable solution.